Constrained Delaunay tetrahedrisation refinement

Internship at the *CNR IMATI* supervised by Marco Attene and Marco Livesu

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ENS of Lyon

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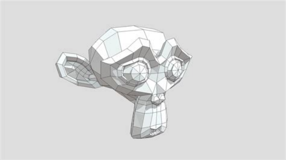
Outline

Big picture

Constrained Delaunay tetrahedrisation



A 3D model
(not necessarly
tetrahedrised)

$\longrightarrow$ The **Delaunay** tetrahedrised model (*i.e.* a set of tetrahedra)

Applications

Physics-based simulations:

- Finite element method
- Solutions of partial differential equations
- Ray tracing

Figure: Example of finite element method

Problem

Need good quality and regularity of the mesh

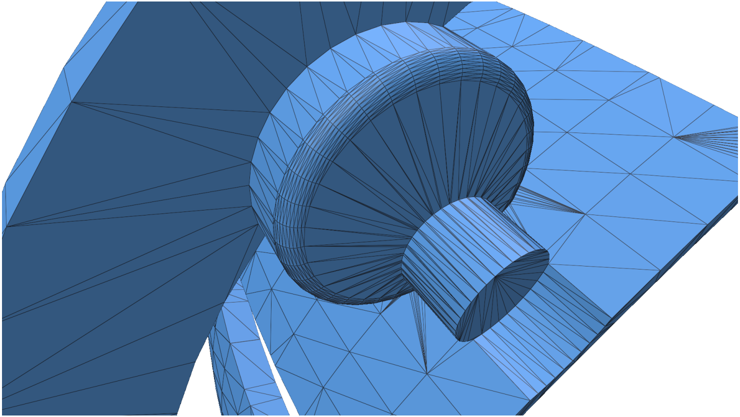


Figure: Example of a bad model

Constrained Delaunay triangulation

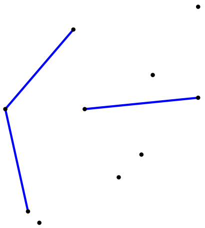
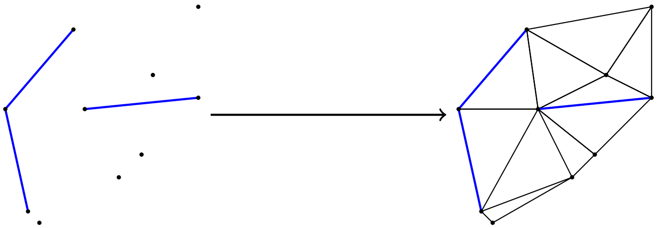


Figure: Edge constraints for Delaunay triangulation

Constrained Delaunay triangulation



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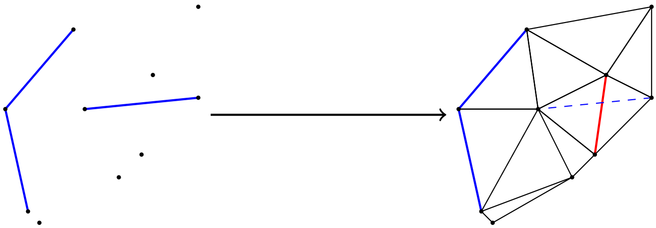

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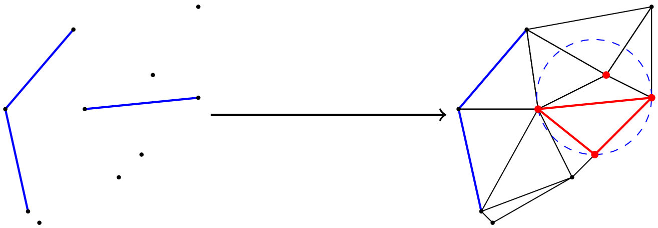

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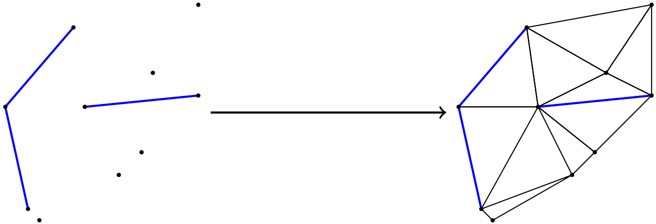

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An edge is either a constraint or is locally Delaunay

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- Triangles become tetrahedra
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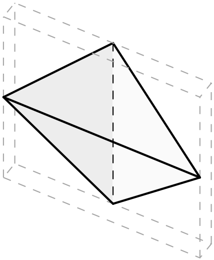

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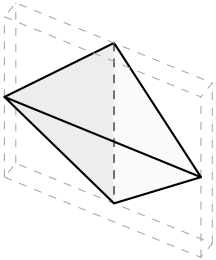

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- Constrained Delaunay tetrahedrisation are not always possible

TetGen [Si, 2015]

Already exisiting software

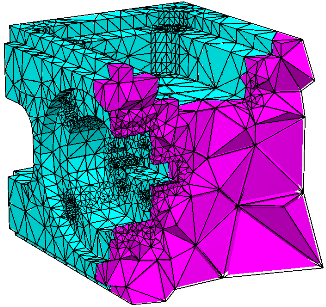
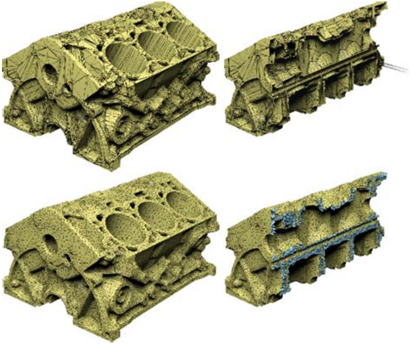


Figure: A model tetrahedrized by TetGen

- Not so robust, crashs on some models

Tetrahedral meshing in the wild (Tetwild) [Hu, 2018]



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- Not strict on boundary

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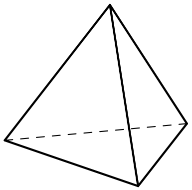

Minimized on a regular tetrahedron ($= 3$)

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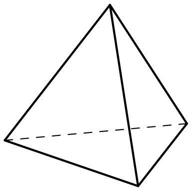

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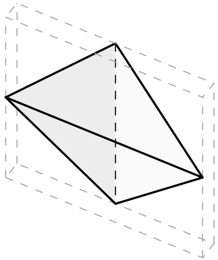
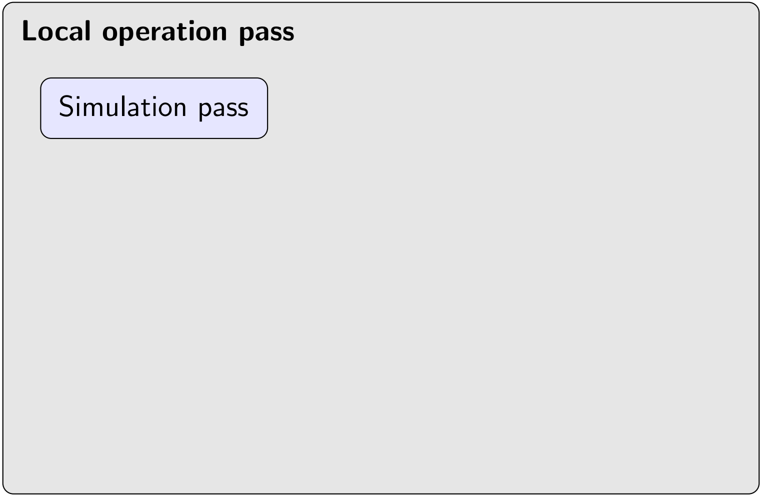


Big value on bad tetrahedron ($\sim 10^3 - 10^4$)

Local operation pass

Figure: Local operation scheme



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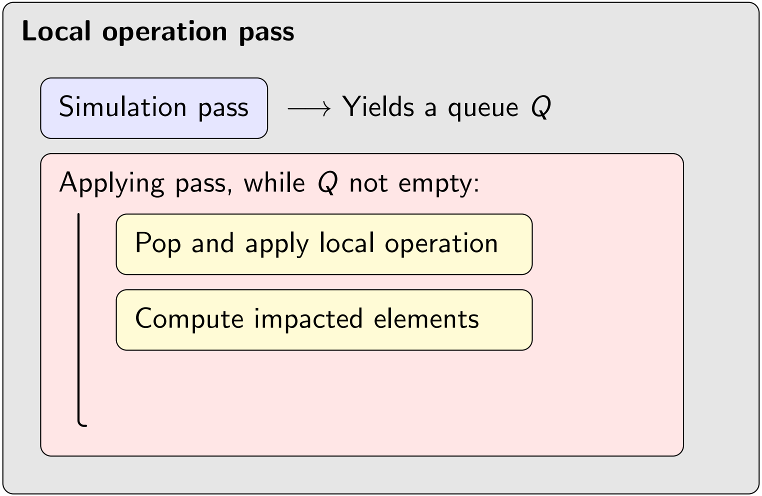

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Local operation pass

Simulation pass $\longrightarrow$ Yields a queue Q

Applying pass, while Q not empty:

- Pop and apply local operation
- Compute impacted elements
- Update Q

Figure: Local operation scheme

Optimization Pass

- Edge split pass
- Edge collapse pass
- Edge swap pass
- Face swap pass
- Vertex smoothing pass
- Tetrahedron collapse pass

Figure: Optimization process scheme

Overview

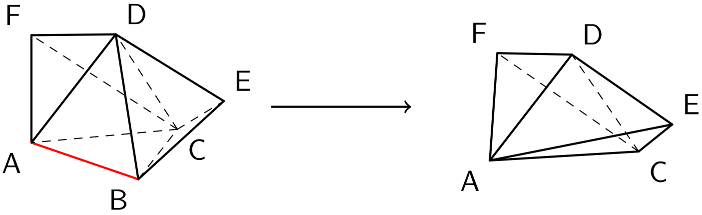

Figure: Example of collapsing an edge

Simulation

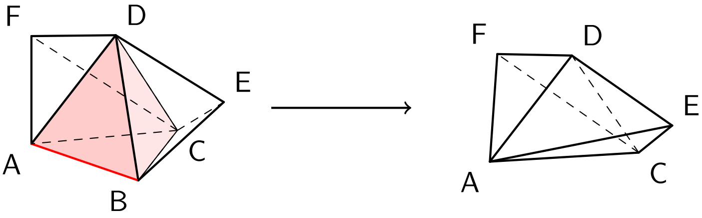

Figure: Impacted tetrahedra by an edge collapse

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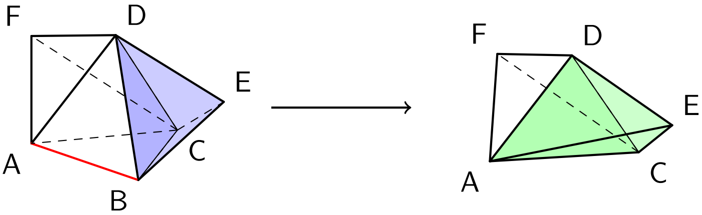

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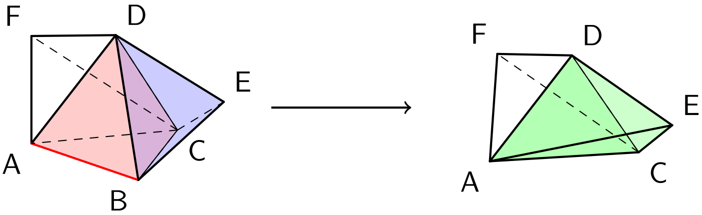

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Quality before:

Max on removed elements $\cup$ unupdated elements

Quality after:

Max on updated elements

Topological failure

Collapsing $e := (u, v)$ can break topology

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$$\mathrm{Lk}(e) = \mathrm{Lk}(u) \cap \mathrm{Lk}(v) \qquad \text{(Link condition)}$$

- $\mathrm{Lk}(u)$ = neighbours of u
- $\mathrm{Lk}(e)$ = vertices sharing a face with e

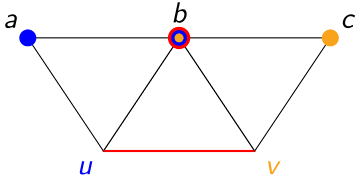

Figure: Respected link condition

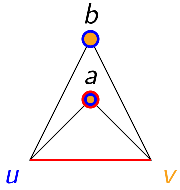

Figure: Not respected link condition

Get rid of slivers

Some slivers resisted known operation $\Rightarrow$ Create a new operation

Figure: Tetrahedron collapsing process

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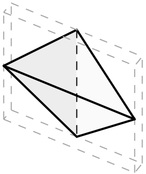

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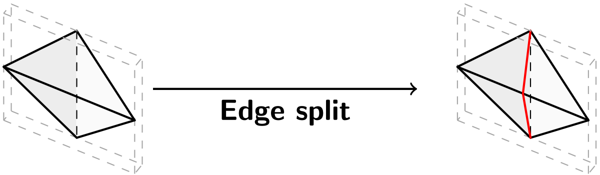

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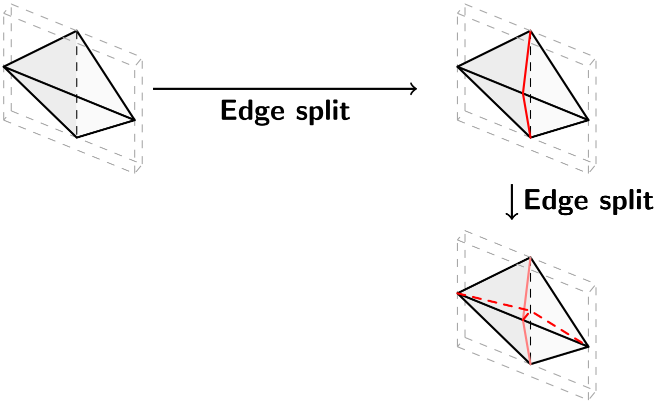

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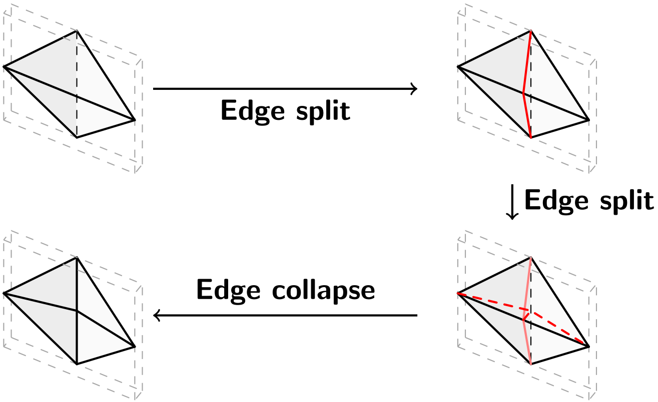

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Comparison

	CDT	TetGen
Fails	0%	25,2%

Table: Stats on the smallest 500 models of *Thingi10k*

- + More robust

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- - Slower

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Best on time	14,44%	85,56%

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Table: Stats on the smallest 500 models of *Thingi10k*

- + More robust
- - Slower
- - Slightly less efficient

	CDT	TetGen
Best on time	14,44%	85,56%
Best on mean	5,35%	94,65%
Best on max	20,59%	79,41%

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Future work and improvements

- Optimize the speed
  - Use more specific data structures

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- Optimize the speed
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- Create new operations
  - To remove needles
- Make some operations finer
  - Split edge not always at midpoint

References

H. Si.
TetGen, a Delaunay-Based Quality Tetrahedral Mesh Generator.
ACM Trans. on Mathematical Software, 41 (2), Article 11, 2015.

Y. Hu, Q. Zhou, X. Gao, A. Jacobson, D. Zorin, D. Panozzo.
Tetrahedral Meshing in the Wild.
ACM Trans. Graph. 37, 4, Article 60, 2018.

Delaunay condition

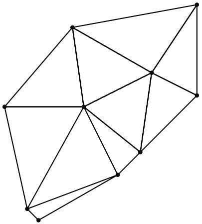

Figure: Example for Delaunay condition

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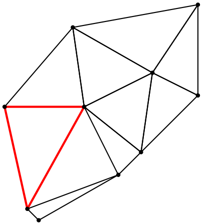

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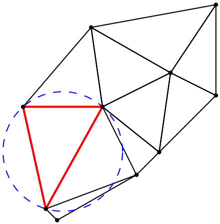

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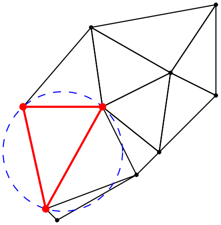

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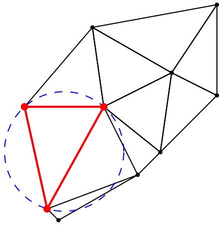

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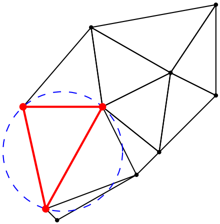

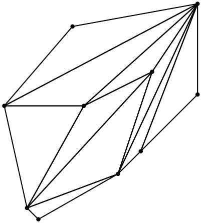

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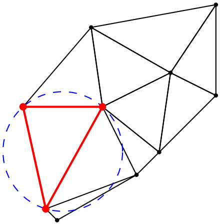

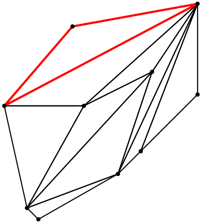

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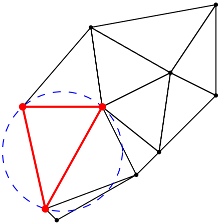

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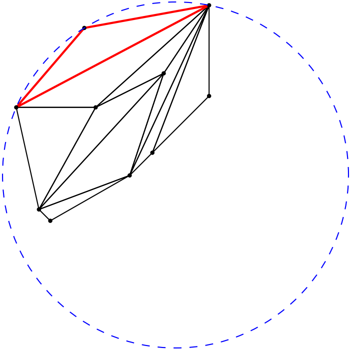

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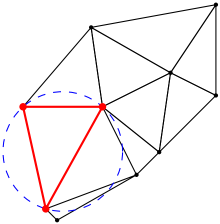

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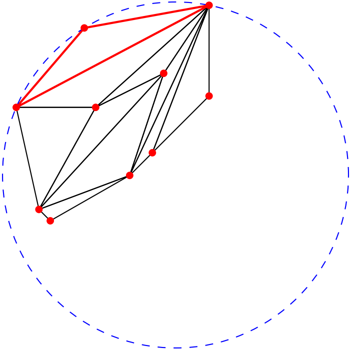

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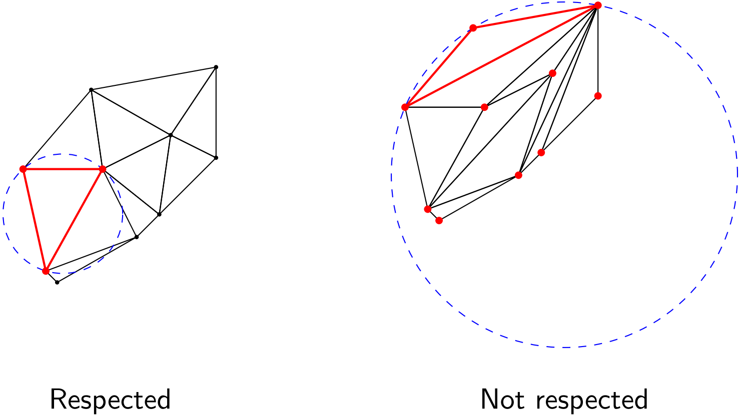

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An example

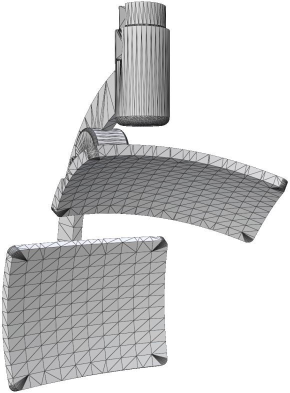
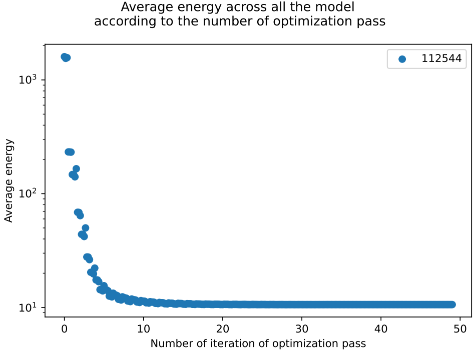


Figure: The model 112544



Evolution of the distribution